



Aetherum

— PRE-SEED · \$1M · 2026

The Crypto Lending Platform That *Explains Itself.*

Crypto-collateralized lending infrastructure for credit unions — with AI-powered underwriting guidance built in.

140M

CU MEMBERS IN
THE US

\$2.2T

CU ASSETS
UNDER
MANAGEMENT

\$0

B2B CRYPTO
LENDING
INFRASTRUCTURE



Credit unions are *locked out* of crypto lending.

140 million Americans trust their credit union over any bank. But when those members want to unlock liquidity from their Bitcoin or Ethereum, they're forced to D2C lenders like Nexo or Ledn — or to sell, triggering a taxable event.

Credit unions want to serve these members. The regulatory complexity is the barrier — not the appetite.

- 01 No compliant crypto lending infrastructure exists for NCUA-regulated institutions.
- 02 D2C competitors (Nexo, Ledn, SALT) compete with credit unions — they don't serve them.
- 03 Younger members with crypto are migrating to fintech — credit unions are losing them.
- 04 No risk assessment model purpose-built for crypto collateral under NCUA guidelines.
- 05 Custody, connectivity, and KYC/AML integration takes months of compliance work per institution.

The *full stack* a credit union needs to go live.

01 · LENDING ENGINE

Crypto-Collateralized Loans

Members keep their BTC/ETH exposure, avoid the taxable sale, and receive USD liquidity at 8–12% APR. Real-time LTV monitoring with automated margin calls.

02 · RISK

DACS™ Risk Assessment

A patent-pending, multi-pillar risk assessment model purpose-built for crypto collateral. AI guidance layer live — members get plain-language borrowing insights from their DACS profile.

PATENT PENDING

03 · CUSTODY

Institutional Custody

Each credit union holds its own direct institutional custody account. Member wallets live within the CU's structure. Aetherum is the software layer only — we never touch member assets.

04 · COMPLIANCE

Compliance Infrastructure

Bank-grade financial data connectivity approved and live across 6 core products. KYC/AML and fraud infrastructure fully executed — production-ready and live.

05 · COMPLIANCE

ERC-3643 On-Chain Identity

Clean-room implementation of three ERC-3643 modules — jurisdiction gating, collateral LTV verification, CU member identity attestation. Built on T-REX, deployed on Sepolia.

PATENT PENDING

06 · COMPLIANCE

ERC-7943 Collateral Token

The first crypto collateral token built to the official Ethereum RWA standard. Every compliance check runs on-chain and is publicly verifiable. No black boxes, no trust-me architecture.

PATENT PENDING

07 · IDENTITY

ONCHAINID

ERC-734 / ERC-735 · Per-CU identity contracts on Base mainnet · CU-owned · KYC claims on-chain · patent pending

PATENT PENDING

08 · YIELD

Staking Module

Members stake supported assets through the platform — earning yield while maintaining collateral eligibility. A new revenue stream for CU partners. Coming soon.

09 · TAX INTELLIGENCE

Tax-Position Score & Optimizer (TPSOE™)

Ranks pledge assets by embedded tax liability so the optimal pledge set is always selected. Members borrow smarter, not just more. No B2B competitor offers this.

PATENT PENDING

10 · MEMBER INTELLIGENCE

Member Crypto Insights

Real-time visibility into the assets members already own — before they need a loan. Consent-based wallet connections and an opt-in qualified-borrower pipeline.

PATENT PENDING

11 · COLLATERAL SURVEILLANCE

Collateral Intelligence Engine (CIE™)

Every position monitored 24/7. Finality-aware on-chain event processing, live LTV in basis points, DACS updated on every price move, institution-authorized liquidation. SOL collateral: live staking APY from Solana RPC — displayed on member dashboard.

PATENT PENDING

12 · REGULATORY

NCUA Examiner View

Read-only audit dashboard scoped per credit union. Live on-chain positions, LTV ratios, liquidation events, and a full BSA/AML audit trail. Built for examination from day one.

13 · SETTLEMENT

Multi-Rail Stablecoin Disbursement

Pluggable multi-rail disbursement supporting USD and SGD across EVM-compatible networks — wired end-to-end into the loan origination and settlement pipeline.

14 · IDENTITY

CU-Owned Identity Layer

Per-credit-union ONCHAINID infrastructure deployed on Base mainnet. CU owns the contracts. Aetherum holds only a revocable claim-signing key. KYC clearance auto-translates to on-chain member eligibility. Zero gas cost via Coinbase Paymaster.

LIVE

15 · AUDIT INFRASTRUCTURE

Arc Transaction Memos

Every disbursement carries structured context on-chain — loan ID, member identifier, branch code — indexed by loan reference, emitted only on success. NCUA examiners verify the audit trail directly from the chain without querying a vendor database.

LIVE

16 · DEBT INSTRUMENTS

CU Debt Instrument

Audit-native tokenized loan participation. Tokenizes pools of member loans as ERC-3643 debt tokens with regulatory-counterparty gating at the transfer layer, policy-gated default via TxGuard's pre-signing engine, and examiner-native audit emission on every lifecycle event.

PATENT PENDING



The *largest untapped* lending opportunity in credit union history.

"Credit unions are sitting on the largest untapped lending opportunity in their history. Aetherum is the infrastructure layer that makes it accessible."

TAM · TOTAL ADDRESSABLE MARKET

\$150B+

Untapped Crypto Lending Gap. 56M Americans hold crypto. Less than 1% can borrow against it through a regulated institution.

SAM · SERVICEABLE MARKET

\$15–25B

NCUA-Regulated Credit Unions. 4,800 CUs, 140M members, ~21M holding crypto — with no compliant path to liquidity.

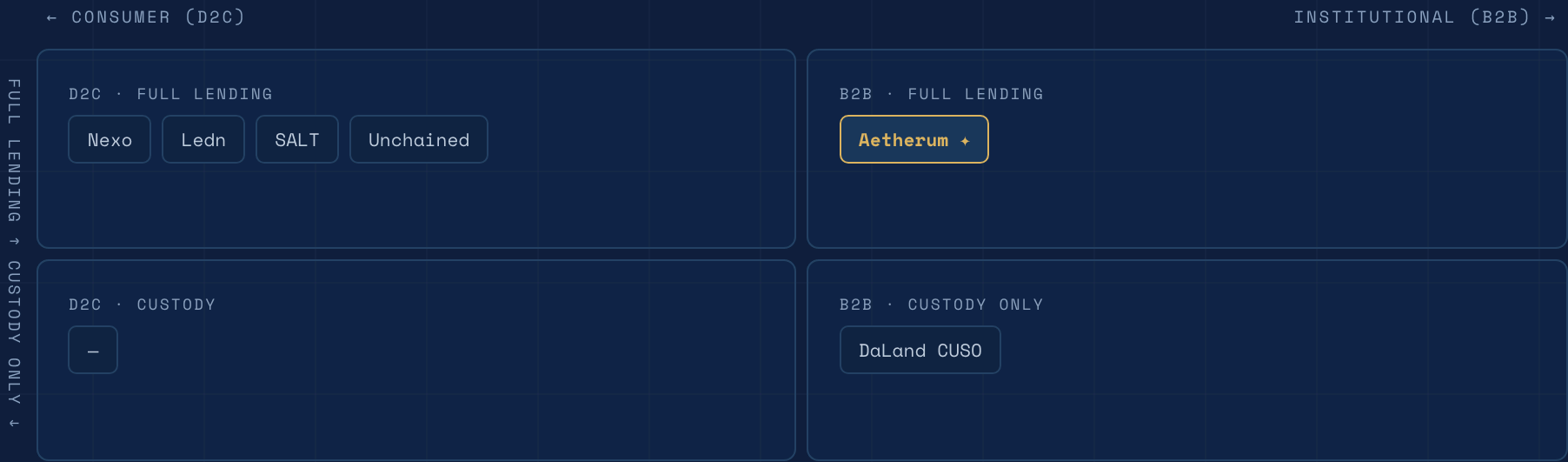
SOM · 3-YEAR RAMP

\$1.2M → \$36M

ARR. Yr 1: 10 CUs × 100 loans/mo. Yr 2: 50 CUs × 150. Yr 3: 150 CUs × 200 = \$36M ARR.



Built for institutions. *No one else is.*



"Every competitor is building faster pipes. Aetherum is building the first crypto lending platform purpose-built for regulated institutions — with the compliance infrastructure to match."



Aetherum

DEFENSIBILITY · FOUR MOAT PILLARS

01 · MOAT

DACS™

Patent-pending multi-pillar risk assessment model. No competitor has a purpose-built crypto risk assessment model for NCUA-regulated institutions.

02 · MOAT

Channel

Board advisor Steve Salzer, SVP at Velera — the credit union technology backbone serving 1,400+ institutions. Channel access that cannot be replicated.

03 · MOAT

IP

21 provisional patent applications covering every core system — risk assessment, identity, tax intelligence, privacy-preserving underwriting, collateral management, on-chain compliance, per-institution owned on-chain identity infrastructure, crypto-collateralized loan-to-card disbursement, pre-signing transaction verification & per-institution policy enforcement, and audit-native tokenized loan participation debt instruments.

04 · MOAT · NEW

Audit-Native

Every decision cryptographically anchored on-chain. NCUA examiners get a verification URL, not a promise. No competitor — DaLand, Nexo, Ledn, Unchained — can say this.



The Entire Industry Is Building Payments Rails. *We're Building the Lending Layer.*

Every major financial institution — Visa, PayPal, Mastercard, Circle, Payoneer — is racing to embed stablecoins into payments, settlement, and cross-border transfers. The entire institutional buildout is pointed at moving money faster.

WHERE THE MARKET IS GOING

Stablecoin payments.

Cross-border settlement.

Merchant acceptance.

Treasury rails.

Consumer wallets.

Source: Q1 2026 public earnings calls — Visa, PayPal, Mastercard, Circle, Payoneer, Remitly, Western Union, d.local, Euronet, Corpay

WHERE AETHERUM IS GOING

Crypto-collateralized lending.

NCUA-compliant infrastructure.

Member identity on-chain.

Audit-native underwriting.

"Credit unions aren't payments companies. They're lenders. Zero players in this chart are building for them — and that's the whitespace Aetherum owns."



Most fintech shows you a dashboard. *Aetherum shows you the reasoning.*

DACS RISK ASSESSMENT

81.78

DACS SCORE

STRONG

VOLATILITY

78.2

LIQUIDITY

84.5

COLLATERAL

83.1

TAX POS.

79.4

REG. RISK

88.0

LTV RATIO

80.6

SMART REVIEW · COMPLIANCE

81.78

DACS SCORE · Approved — \$24,500

KYC Identity Verified

PASSED

Sanctions & AML Screening

CLEAR

ERC-3643 On-Chain Attestation

PASSED

US Jurisdiction Verified

CONFIRMED

TPSOE™ Tax Efficiency Score

74 / 100

"Based on your DACS profile, pledging ETH minimises your tax exposure by ~\$3,200 vs. selling. Recommended borrow: \$24,500 at 65% LTV."

LOAN OFFICER DASHBOARD

12

PENDING

\$1.4M

VOLUME MTD

96%

COMPLIANCE

MEMBER	AMOUNT	DACS	STATUS
A. Reyes	\$24,500	81.8	APPROVED
M. Chen	\$50,000	77.2	REVIEW
T. Williams	\$12,000	62.4	PENDING
S. Patel	\$95,000	88.1	APPROVED
D. Kim	\$8,500	41.9	FLAGGED

DACS™ Risk Gauge · Smart Review™ · Loan Officer Portal — every loan comes with an explanation the member can read, and a record the examiner wants to see.



Built to the *compliance bar* that keeps others out.

ARC MEMOS	Live on testnet — On-chain audit trail wired into disbursement flow. Loan ID, member ID, and branch code travel with every USDC transfer. Indexed on-chain, examiner-readable without vendor database access.	LIVE ON TESTNET
NIUM CARD	Sandbox live — Full borrow → disburse → spend loop closed. Member borrows against crypto, spends via Visa card. End-to-end on Circle rails.	LIVE
BANK CONNECTIVITY	Financial data connectivity (CRA bundle) approved and live — 6 core products active.	LIVE
SARDINE	KYC/AML, fraud detection & transaction monitoring — Sardine MSA executed. Real-time OFAC/sanctions screening, device intelligence, behavioral fraud signals, and BSA/AML compliance. Consortium anonymization model provides cross-institution fraud signals without exposing member data.	LIVE
3 CORE SYSTEMS	Jack Henry FIN contract executed (Banno live in Garden sandbox). Corelation KeyStone certified. Fiserv AppMarket in progress.	LIVE
CUSTODY	Enterprise custody live — each CU holds its own direct account. Non-custodial by architecture.	LIVE
LIQUIDATION	Three-tier liquidation escalation with human oversight for every large position. Full audit trail.	LIVE
STABLECOIN SETTLEMENT	Multi-rail disbursement live in production — USD & SGD across EVM networks; cross-chain transfer + programmable wallets operational.	LIVE
RIPPLE / XRPL	XLS-66 LoanBroker live on testnet — dual-signature LoanSet flow operational. RLUSD as native loan currency.	LIVE ON TESTNET
ARC · CIRCLE	Arc — The Economic OS · Stablecoin-native L1. USDC native gas. Sub-second finality. Native Circle-stack integration. AetherumCollateralEscrow deployed to Arc Testnet. Milestone 2 settlement infrastructure. Mainnet Q3 2026.	LIVE
CIRCLE OCC	ECOSYSTEM NOTE Circle National Trust Bank — OCC preliminary conditional approval (First National Digital Currency Bank, N.A.) — July 2026.	APPROVED
PHAROS	USDC + CCTP integration live on Pharos Atlantic Testnet.	LIVE
MESH	Collateral Transfer Rail · MSA Executed. Sandbox \$25K USDC transfer verified.	LIVE
CARD FUNDING BRIDGE	Nium sandbox live, 200 OK confirmed. Rain MNDA signed. Borrow→disburse→spend loop complete. White-label virtual card issuance triggered by stablecoin disbursement. Patent pending.	LIVE
NATIVE LTV MONITOR	BTC/ETH collateral polled every 60 seconds — 30-day rolling volatility feeds DACS. No third-party dependency.	LIVE
PLATFORM	Full-stack MVP live at test-app.aetherum.ai. ERC-3643 modules deployed on Sepolia.	LIVE



DACS™	Ten-pillar risk assessment model, live gauge wired to API.	PATENT PENDING
TPSOE™	Tax-Position Score & Collateral Optimizer ranks pledge assets by embedded tax liability. No B2B competitor offers this.	PATENT PENDING
FHE-DACS	Homomorphic encryption layer — member data stays encrypted throughout underwriting.	PATENT PENDING
CIE™	Real-time collateral surveillance pipeline live in production.	PATENT PENDING
ERC-3643 / 7943	Only on-chain compliance layer purpose-built for CU lending. Deployed and verified on Sepolia (T-REX standard).	LIVE ON TESTNET
COMPLIANCE RECEIPTS	On-chain compliance receipt system live on Sepolia. SHA-256 commitment per lending decision; examiners get a verification URL.	PATENT PENDING
DACS EXPLAINABILITY	Examiner-grade underwriting records with factor-level breakdowns & 12 CFR Part 723 citation. PDF export for NCUA submission.	PATENT PENDING
SUPERVISORY CONTROL	NCUA-aligned supervisory hold system with on-chain attestation and CSV examination export.	PATENT PENDING
SESSION-ANCHORED CREDIT	Session-Anchored Collateral Credit System with Audit-Native Receipt Generation — Patent-pending architecture for session-anchored collateral credit origination with integrated audit-native receipt generation.	PATENT PENDING
DDQ ENGINE	387-entry security questionnaire (full SIG framework). A CU's risk team can complete a full vendor security review without a call. Live at aetherum.ai/ddq.	LIVE
SMART REVIEW	AI-powered underwriting guidance live in production — zero added friction in the loan flow.	LIVE
ECOSYSTEM ADVISOR	Steve Salzer, SVP at Veleria — Board Advisor. NACUSO Regulatory Advocacy Committee & CUNA Payments Subcommittee.	ACTIVE
ON-CHAIN IDENTITY	Per-CU ClaimIssuer + IdFactory deployed on Base. CU-owned contracts. CDP Paymaster gas sponsorship live. Every identity event generates a tamper-evident audit receipt.	LIVE
CU IDENTITY INFRASTRUCTURE	Per-Institution Owned On-Chain Identity Infrastructure with Escrow-Transfer Ownership and Scoped Revocable Vendor Signing Authority for Regulated Financial Institutions.	PATENT PENDING
LOAN-TO-CARD DISBURSEMENT	Crypto-Collateralized Loan-to-Card Disbursement with Audit-Native Compliance Infrastructure for Federally Regulated Depository Institutions — patent-pending architecture for borrow→disburse→spend loop with white-label card issuance.	PATENT PENDING
TXGUARD	AetherumTxGuard — Pre-Signing Transaction Verification & Per-Institution Policy Enforcement — patent-pending system and method for pre-signing transaction verification, per-institution policy enforcement, and audit-native compliance logging for digital asset operations in NCUA-regulated credit union lending infrastructure.	PATENT PENDING
CU DEBT INSTRUMENT	Audit-Native Tokenized Loan Participation — patent-pending architecture for tokenized loan participation instruments with regulatory-counterparty transfer gating, policy-gated collateral liquidation via pre-signing policy engine, and examiner-native audit emission for NCUA-regulated credit union lending infrastructure.	PATENT PENDING
IP / PATENTS	21 provisional patent applications filed with the USPTO across the full platform.	21 PATENTS PENDING



Six streams. *Platform-fee led.*

ARR TARGET

\$1,177/loan × 100 loans/mo × 10 CUs = \$1.2M ARR

60%

Platform Fees

\$500–\$1,500 per loan originated. Primary revenue driver, tied to volume.

18%

Transaction Fees

0.5–1% on crypto collateral transactions processed through the platform.

7%

SaaS License

\$50K–\$150K/year for large credit unions. Predictable ARR layer at scale.

15%

Staking Yield Fee

15bps on total staking yield via YieldGuardService. Patent-protected (AETHERUM-018) — turns idle collateral into a third income stream.

New

Tax Intelligence Module

\$10K–\$25K/year per CU as a premium add-on. No B2B competitor offers this to institutions.

New

ERC-3643 + ERC-7943 Compliance Layer

Per-attestation module licensing. Any RWA issuer requiring US jurisdiction or CU member identity verification licenses our modules.

"Staking transforms Aetherum from a loan-fee business into a yield-management infrastructure platform. Every dollar of collateral in custody earns for three parties simultaneously — the CU, the member, and Aetherum."



Built by operators. *Advised by the ecosystem.*

John Bailey

FOUNDER & CEO

20+ years in technical recruiting with placements at Google, Spotify, and Goldman Sachs. Previously co-founded Thorium Data. Building Aetherum full-time since early 2025 — product, BD, fundraising, and technical implementation.

Steve Salzer

BOARD ADVISOR · SVP, VELERA

SVP at Velera since 2017. Chairs the Board of Managers at Ongoing Operations LLC. NACUSO Regulatory Advocacy Committee and CUNA Payments Subcommittee. JD, University of Chicago Law School.

Rahul Goel

ADVISOR · JP MORGAN / UBER

Cybersecurity and financial infrastructure expertise from senior roles at JP Morgan and Uber, including involvement with Meta's Novi digital wallet. Advises on platform security and institutional trust.

Saiprasad Raut

ADVISOR · WORLDPAY / VANAR

Head of Payments Infrastructure at Vanar, formerly Worldpay. Deep payments and blockchain infrastructure expertise guiding settlement and custody architecture.



Closing a *\$1M Pre-Seed* Round.

We're in the final stretch of closing our pre-seed round. Proceeds go toward closing the first two credit union pilots, completing platform development, and bringing on a CTO. Capital deploys us to the first 3 credit union pilots — reducing market risk and proving the explanation layer at scale.

ROUND SIZE	\$1,000,000
INSTRUMENT	Post-Money SAFE
VALUATION CAP	\$8M Post-Money
DISCOUNT	20%
TIMELINE	Closing Soon

Let's talk.

20 minutes. We'll walk through the platform live and answer any diligence questions.

BOOK A CALL